



MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

MALAWI INSTITUTE OF TECHNOLOGY

APPLIED STUDIES DEPARTMENT

MID SEMESTER EXAMINATIONS

PHYS-111: MECHANICS AND THERMAL PROPERTIES OF MATTER

[40 marks]

Examiners: F. Benard, R. Chimombo, S. Chisale, C. Malanda & P. Mnjeza.

DATE: 5th MAY 2021

TIME: 2hrs: 30min(17:00am - 19:30pm)

INSTRUCTIONS

1. Write your **Examination Number** on each page of your answer sheets.
 2. Start each question on a fresh page
 3. This paper contains **3 pages** and **4 QUESTIONS**, please check.
 4. Answer **ALL** questions.
 5. Marks allocated for each question are indicated against it.
 6. Submit your answer sheet physically to your respective lecturers.
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LIST OF PHYSICAL CONSTANTS

Acceleration due to gravity	g	9.8 m/s^2
Universal gravitation constant	G	$6.67 \times 10^{-11} \text{ N.m}^2/\text{kg}^2$

Question 1.

- The position of a particle in space is described by (3.00, 4.00) m. Find the polar coordinates of this particle. **[4 marks]**
- The displacement (x) of an object is directly proportional to its acceleration (a) raised to some power n and the time taken (t) raised to some power m . Find the simplest form of equation for the displacement. **[6 marks]**

Question 2.

In Mangochi, Malawi, the ultimate test of a young man's courage once was to jump off a bridge into the Shire River, 25.0 m below the bridge.

- How long did the jump last? **[4 marks]**
- How fast was the diver traveling upon impact with the water? **[3 marks]**

Question 3.

- A car travels from A to B along 30 km north, then 60 km east, then 110 km south as shown in **figure 1**. Determine the displacement of the car from A to D. **[6 marks]**

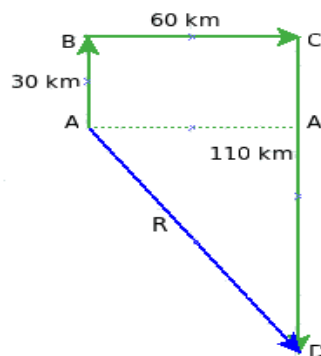


Figure 1

- b. During a training session at Juventus, Cristiano Ronaldo practices free kicks by shooting a ball towards a 2.4 m tall goal from a distance of 30 m. He kicks a ball at an angle of 10° above the ground with the velocity of 25 m/s, assuming the goal is empty (no goalkeeper), will he make a goal? [7 marks]

Question 4

- a. State Newton's first law of motion. [2 marks]
- b. Find the tension in each cord in **Figure 2** if the weight of the suspended object is 250 N. [8 marks]

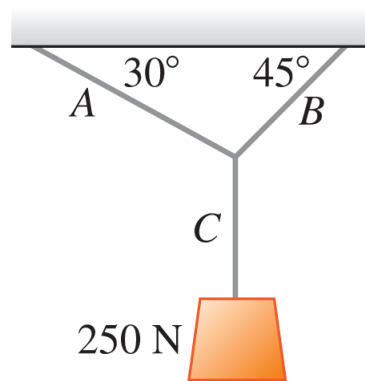


Figure 2

END OF EXAMINATION PAPER